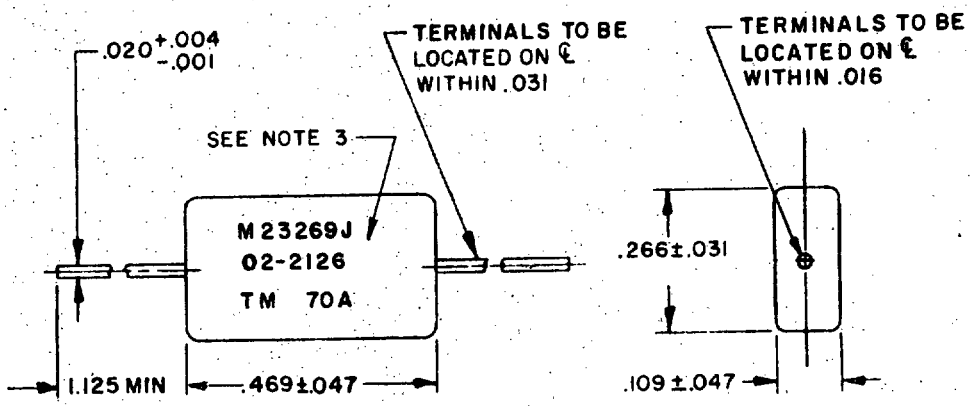
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, GLASS DIELECTRIC, (AXIAL WIRE-LEAD TERMINALS), ESTABLISHED RELIABILITY, STYLE CYR15

(E) FAILURE RATE LEVEL 'L' IS INACTIVE FOR DESIGN AFTER 20 July 1983.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-23269.



INCHES	MM
.001	.03
.004	.10
.016	.41
.020	.51
.031	.79
.047	1.19
.109	2.77
.266	6.76
.469	11.91
1.125	28.58

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. See marking requirement.

FIGURE 1. Dimensions and configurations.

(E) denotes change.

FSC 5910

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case material: Glass.

Terminals: Axial wire-lead; solder coated, copper-clad, nickel-iron core (46 percent nickel, 54 percent iron).

Capacitance (nominal): See table I.

Capacitance tolerance: See table I.

Rated voltage: See table I.

Failure rate level: L, M, P, R, or S.

Rated temperature: -55°C to +125°C.

Temperature coefficient: 140 ±25 parts/million/°C.

Capacitance drift: 0.1 percent or 0.1 pF, whichever is greater.

Seal: Hermetic.

Seal: Test I.

Insulation resistance: 10,000 megohms or 200 ohm-farads minimum (qualification inspection only at +125°C). 100,000 megohms minimum (at +25°C).

Dissipation factor (DF): 0.1 percent maximum.

Thermal shock and moisture resistance:

DF: 0.1 percent maximum.

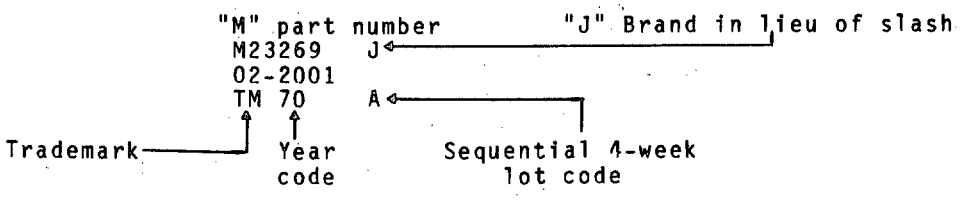
Quality factor (Q): In accordance with MIL-C-23269.

Life:

ΔCapacitance (240 hour and 2,000 hour): 0.5 percent maximum.

DF (240 hour and 2,000 hour): 0.15 percent maximum.

Marking:



Substitutability data: See table II.

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TABLE I. Electrical and physical characteristics and dash numbers.

Capacitance	Capacitance tolerance	Dash number M23269/02- FR level in percent/1,000 hours					
		500 V dc					100 V dc
		L (2.0)	M (1.0)	P (0.1)	R(0.01)	S (0.001)	S (0.001)1/
(pF)							
220	±1%	2001	3001	4001	5001	6001	7001
220	±2%	2002	3002	4002	5002	6002	7002
220	±5%	2003	3003	4003	5003	6003	7003
240	±1%	2004	3004	4004	5004	6004	7004
240	±2%	2005	3005	4005	5005	6005	7005
240	±5%	2006	3006	4006	5006	6006	7006
270	±1%	2007	3007	4007	5007	6007	7007
270	±2%	2008	3008	4008	5008	6008	7008
270	±5%	2009	3009	4009	5009	6009	7009
300	±1%	2010	3010	4010	5010	6010	7010
300	±2%	2011	3011	4011	5011	6011	7011
300	±5%	2012	3012	4012	5012	6012	7012
330	±1%	2013	3013	4013	5013	6013	7013
330	±2%	2014	3014	4014	5014	6014	7014
330	±5%	2015	3015	4015	5015	6015	7015
360	±1%	2016	3016	4016	5016	6016	7016
360	±2%	2017	3017	4017	5017	6017	7017
360	±5%	2018	3018	4018	5018	6018	7018
390	±1%	2019	3019	4019	5019	6019	7019
390	±2%	2020	3020	4020	5020	6020	7020
390	±5%	2021	3021	4021	5021	6021	7021
430	±1%	2022	3022	4022	5022	6022	7022
430	±2%	2023	3023	4023	5023	6023	7023
430	±5%	2024	3024	4024	5024	6024	7024
470	±1%	2025	3025	4025	5025	6025	7025
470	±2%	2026	3026	4026	5026	6026	7026
470	±5%	2027	3027	4027	5027	6027	7027
510	±1%	2028	3028	4028	5028	6028	7028
510	±2%	2029	3029	4029	5029	6029	7029
510	±5%	2030	3030	4030	5030	6030	7030
		300 V dc					100 V dc
560	±1%	2031	3031	4031	5031	6031	7031
560	±2%	2032	3032	4032	5032	6032	7032
560	±5%	2033	3033	4033	5033	6033	7033
620	±1%	2034	3034	4034	5034	6034	7034
620	±2%	2035	3035	4035	5035	6035	7035
620	±5%	2036	3036	4036	5036	6036	7036
680	±1%	2037	3037	4037	5037	6037	7037
680	±2%	2038	3038	4038	5038	6038	7038
680	±5%	2039	3039	4039	5039	6039	7039
750	±1%	2040	3040	4040	5040	6040	7040
750	±2%	2041	3041	4041	5041	6041	7041
750	±5%	2042	3042	4042	5042	6042	7042
820	±1%	2043	3043	4043	5043	6043	7043
820	±2%	2044	3044	4044	5044	6044	7044
820	±5%	2045	3045	4045	5045	6045	7045
910	±1%	2046	3046	4046	5046	6046	7046
910	±2%	2047	3047	4047	5047	6047	7047
910	±5%	2048	3048	4048	5048	6048	7048
1,000	±1%	2049	3049	4049	5049	6049	7049
1,000	±2%	2050	3050	4050	5050	6050	7050
1,000	±5%	2051	3051	4051	5051	6051	7051

See footnote at end of table.

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TABLE I. Electrical and physical characteristics and dash numbers - Continued.

Capacitance	Capacitance tolerance	Dash number M23269/02- FR level in percent/1,000 hours					
		300 V dc					100 V dc
		L (2.0)	M (1.0)	P (0.1)	R(0.01)	S (0.001)	S (0.001)1/
(pF)							
1,100	±1%	2052	3052	4052	5052	6052	7052
1,100	±2%	2053	3053	4053	5053	6053	7053
1,100	±5%	2054	3054	4054	5054	6054	7054
1,200	±1%	2055	3055	4055	5055	6055	7055
1,200	±2%	2056	3056	4056	5056	6056	7056
1,200	±5%	2057	3057	4057	5057	6057	7057

1/ Sixth power rule applies for derating at 0.5 $\frac{\text{operating}}{\text{rated}}$ voltage ratios and below.

TABLE II. Substitutability data.

Superseded type designation in MIL-C-23269/2B	Use dash number M23269/02- FR level in percent/1,000 hours		Superseded type designation in MIL-C-23269/2B	Use dash number M23269/02- FR level in percent/1,000 hours	
	L (2.0)	M (1.0)		L (2.0)	M (1.0)
CYR15C221F	2001	3001	CYR15C511J	2030	3030
CYR15C221G	2002	3002	CYR15C561F	2031	3031
CYR15C221J	2003	3003	CYR15C561G	2032	3032
CYR15C241F	2004	3004	CYR15C561J	2033	3033
CYR15C241G	2005	3005	CYR15C621F	2034	3034
CYR15C241J	2006	3006	CYR15C621G	2035	3035
CYR15C271F	2007	3007	CYR15C621J	2036	3036
CYR15C271G	2008	3008	CYR15C681F	2037	3037
CYR15C271J	2009	3009	CYR15C681G	2038	3038
CYR15C301F	2010	3010	CYR15C681J	2039	3039
CYR15C301G	2011	3011	CYR15C751F	2040	3040
CYR15C301J	2012	3012	CYR15C751G	2041	3041
CYR15C331F	2013	3013	CYR15C751J	2042	3042
CYR15C331G	2014	3014	CYR15C821F	2043	3043
CYR15C331J	2015	3015	CYR15C821G	2044	3044
CYR15C361F	2016	3016	CYR15C821J	2045	3045
CYR15C361G	2017	3017	CYR15C911F	2046	3046
CYR15C361J	2018	3018	CYR15C911G	2047	3047
CYR15C391F	2019	3019	CYR15C911J	2048	3048
CYR15C391G	2020	3020	CYR15C102F	2049	3049
CYR15C391J	2021	3021	CYR15C102G	2050	3050
CYR15C431F	2022	3022	CYR15C102J	2051	3051
CYR15C431G	2023	3023	CYR15C112F	2052	3052
CYR15C431J	2024	3024	CYR15C112G	2053	3053
CYR15C471F	2025	3025	CYR15C112J	2054	3054
CYR15C471G	2026	3026	CYR15C122F	2055	3055
CYR15C471J	2027	3027	CYR15C122G	2056	3056
CYR15C511F	2028	3028	CYR15C122J	2057	3057
CYR15C511G	2029	3029			

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Air Force - 11
NASA - NA

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Air Force - 17, 85
DLA - ES

User activities:
Navy - MC, CG, AS, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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